

# JOONYOUNG JEONG

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## RESEARCH INTERESTS

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My research interests lie in cellular network security (5G RAN and 5GC) and computer systems security. I also enjoy dissecting binaries to uncover vulnerabilities and analyze malware functionality.

## EDUCATION

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**Iowa State University** Aug 2021 – Jul 2023

M.S. in Electrical and Computer Engineering

GPA: 3.95/4.0

*Relevant coursework: Communication Systems, Advanced Communications, Network Protocol Security, Cryptography, Data Analytics in ECE, Digital Signal Processing, Introduction to Deep Learning.*

**Korea Military Academy** Mar 2009 – Mar 2013

B.E. in Mechanical Engineering

## RESEARCH EXPERIENCE

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**Mobile Security Architecture for Defense 5G Infrastructure** Jul 2026 – Dec 2026

*Sponsored Research funded by ROK Army Headquarters*

*Analyzing defense mobile system architectures and domestic/international security policies (US DoD, NATO, 3GPP) to identify threat vectors relevant to 5G military network deployment.*

**Flow Matching for Missing Data Imputation [1]** Mar 2026 – Present

*Collaborative Research Project*

**Cost-Aware UE-Side Mobility Management for TN-NTN Networks [2]** Sep 2025 – Aug 2026

*Independent research; single-author paper accepted to ACM/IEEE MSWiM 2026*

**Covert authentication for securing MIMO communications [3]** Jan 2022 – Jul 2023

*Supervisor: Dr. Sang Kim, Iowa State University*

**Research Assistant, Iowa State University** Aug 2022 – Jul 2023

*Summarized 5G-related papers and 5GC/RAN architectures into advisor briefs*

## PROFESSIONAL EXPERIENCE

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**Rotating Faculty, Korea Military Academy** Jul 2025 – Present

*Teaching and advising cadets; participating in defense projects.*

**Planning Officer, AI R&D Center, Korea Military Academy** Jul 2026 – Present

*Research planning and coordination for AI-driven defense technology initiatives.*

**Signals Intelligence Officer, Ministry of National Defense** Aug 2023 – May 2025

*Malware Analysis and Cyber Threat Intelligence*

**Signals Intelligence Officer, Ministry of National Defense** Aug 2018 – Jul 2021

*COMINT*

**Military Intelligence Officer, Army Headquarters** Mar 2013 – Aug 2018

*General Military Intelligence*

## TEACHING

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**Rotating Faculty, Korea Military Academy**  
*Introduction to Statistics (Fall 2025, Spring 2026)*  
*Introduction to Data Mining (Fall 2025)*  
*Linear Algebra (Spring 2026)*

Jul 2025 – Present

**Teaching Assistant, Iowa State University**  
*Communication Systems II Lab (Spring 2023)*  
*Communication Systems I Lab (Fall 2022)*

Aug 2022 – Jul 2023

## SKILLS

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**Programming Languages:** Python, C/C++, x86-64 Assembly, MATLAB, Simulink, R  
**Analysis Tools:** Burp Suite, Ghidra, IDA Pro, x64dbg, GDB, Wireshark, AFL++

## HONORS & AWARDS

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**Chairman of the Joint Chiefs of Staff Commendation**  
*Joint Chiefs of Staff, Republic of Korea*

2024

**Letter of Appreciation (SES-2)**  
*Foreign Agency*

2024

**Army Merit Scholarship (Full tuition and stipend)**  
*Ministry of National Defense*

Aug 2021 – Jul 2023

## PROJECTS

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**Intrusion Detection using Denoising Autoencoder and Deep Learning**  
*Designed a DNN classifier using a stacked denoising autoencoder for feature learning*

2023

## PUBLICATIONS & WORKSHOP PAPERS

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- [1] J. Jeong, S. Hwang, and Y. Kwon, “Missflow: Flow matching for uncertainty-quantified imputation of missing tabular data,” in *EIML@ICML 2026 Workshop*, Accepted; revisions in progress, 2026.
- [2] J. Jeong, “CALM: UE-side mobility management for integrated TN-NTN networks,” in *ACM/IEEE Int. Conf. on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, Accepted; to appear, Paris, France, 2026.
- [3] J. Jeong, “Covert authentication for securing MIMO communications,” M.S. thesis, Iowa State University, Ames, IA, USA, 2023. [Online]. Available: <https://dr.lib.iastate.edu/bitstreams/d0ca9bbb-4436-4681-97b7-c97fa3a4ec05/download>.